

Unsupervised SEC Form 4 Footnote Change Detection using Fine-Tuned Sentence Transformer and Multi-Task LSTM Networks

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Abstract. Insiders of public companies, including Officers, Directors, and Owners of 10% or more of the company, are required to disclose their insider trading activity with the Securities and Exchange Commission (SEC) through Form 4 filings. Insiders typically include boilerplate footnote language in their filings to avoid extra scrutiny, but underlying shifts in this language may signal changes in an insider’s near-term sentiment regarding their company’s future stock performance. This study introduces a predictive embedding framework that identifies material deviations in insider sentiment preceding abnormal returns. Unlike prior approaches that compare semantic similarity between discrete time periods, this work employs a time-series multi-task LSTM neural network that predicts expected footnote embeddings based on historical trading patterns, identifying material deviations when actual embeddings differ from predictions. A fine-tuned sentence transformer model generates domain-specific embeddings, and material differences are analyzed using an event study framework to measure Cumulative Average Abnormal Returns (CAAR). The approach processes over 3.3 million Form 4 filings from 2006 through 2025. Results demonstrate the effectiveness of deep learning approaches in identifying information-driven trading behavior from textual disclosures, with material changes showing -10.9% lower CAAR than non-material footnote changes at longer sequencing, providing empirical evidence that the model captures meaningful trading signals not immediately reflected in market price.

Keywords: Long Short-Term Memory (LSTM) · Anomaly Detection · Insider Trading · SEC Form 4 · Sentence Embeddings · Multi-Task Learning

1 Introduction

Corporate insiders possess material non-public information, and their trades often signal future performance [2]. SEC Form 4 filings [17] disclose these transactions, providing critical context in unstructured footnotes. Despite evidence that semantic changes in corporate disclosures contain predictive information [3,20], these footnotes remain largely underutilized. While insiders often use boilerplate language, deviations may signal shifts in sentiment. This study investigates whether such shifts can be automatically detected to identify information-driven trading behavior.

Table 1. An example of SEC Form 4 footnote change detection using sequence analysis. The table lists a single issuer-insider trading sequence with nine transactions, of which the footnotes of the last (9th) transaction was changed.

#	Date	Total Shares	Total Value	Days Since Prior Sale	Days Filing Was Late	10b5-1	Related Footnote Shares	Footnote Text
1	2021-05-19	6,367	\$168,980	1	2	No	0	This transaction was executed in...
2	2021-05-20	9,290	\$250,830	1	2	No	0	[NOF]
...								
8	2021-12-09	11,787	\$272,043	191	4	No	11,787	This transaction was executed in...
9	2022-07-21	83,373	\$2,793,029	224	0	No	0	Disposed of as a result of the ...

Table 1 illustrates the approach: the first 8 transactions predict the footnote embedding for the 9th. In the example (SEC accession 0000878526-22-000084), the 9th transaction represents a merger (“Disposed of as a result of the Merger...”) rather than a routine sale, yielding cosine similarity of 0.516—in the bottom 15% of the distribution. Unlike prior approaches comparing discrete time periods [3], the multi-task LSTM learns an insider’s historical patterns and predicts the expected footnote embedding, capturing temporal evolution. Event studies confirm these material deviations are associated with lower subsequent returns. The approach processes over 3.3 million Form 4 filings (2006–2025), using a fine-tuned sentence transformer [16] and a multi-task LSTM [21] evaluated via standard event study methods [6].

2 Related Work

Prior studies establish that insider transactions encode information that can forecast future stock performance [2], with top executive trades associated with significant abnormal returns. Research [10] examines how insiders use SEC Rule 10b5-1 trading plans strategically, with different return patterns than discretionary trades. All 10b5-1 transactions are disclosed in Form 4 footnotes. Machine learning techniques have also been applied to identify patterns and predict illegal insider trading events [19].

Lazy Prices [3] established that semantic changes in corporate disclosures (10-K and 10-Q) exhibit forecasting power for subsequent returns. This aligns with the broader rise of alternative data in finance, where non-traditional sources like search trends [4] provide early market signals. Recent advances in Large Language Models (LLMs) and specialized transformers like FinBERT [12] have further demonstrated that deep semantic analysis can capture market-moving sentiment more effectively than traditional frequency-based metrics [13]. While these studies primarily focus on periodic news or quarterly earnings, our focus on high-frequency SEC Form 4 footnotes provides a novel semantic signal capturing insider sentiment shifts.

Sentence-BERT [16] uses Siamese BERT networks to generate sentence embeddings. While general-purpose embeddings are widespread, domain-specific tuning on small models can be as performant [8]. This study fine-tunes a sentence transformer model on Form 4 footnote text to generate domain-specific embeddings.

Deep learning approaches are effective for analyzing time series data in financial domains [5,21]. This study constructs a multi-task LSTM neural network using hard

parameter sharing, where all predictions share a single LSTM encoder with task-specific prediction heads.

Recent research has applied machine learning directly to Form 4 filings: random forest models achieved over 95% accuracy [15] in detecting illegal insider trading using 110 transaction features, and MaBoost achieved an F1-Score over 94% for classifying high-risk late-reported trades [9]. Neither study uses transaction footnotes or attempts to capture semantic sentiment. This study focuses on Form 4 footnote disclosures, leveraging a multi-task LSTM model to identify deviations from expected footnote patterns and test whether they contain predictive signals not immediately reflected in market prices.

3 Methodology

3.1 Overview and Contribution

This study employs a two-stage methodology: fine-tuning a sentence transformer model on domain-specific footnote text to generate semantic embeddings (vector representations of footnote text), and using a multi-task LSTM neural network to predict expected footnote embeddings based on historical transaction patterns. Material differences are identified when actual embeddings deviate substantially from predicted embeddings, measured by cosine similarity, and evaluated through event study analysis to assess their association with subsequent abnormal stock returns.

Our work differs from prior approaches in several respects. While prior machine learning studies of insider trading [9,15] rely exclusively on structured transaction features, this study leverages unstructured Form 4 footnote text appearing in 19 locations throughout Form 4 filings to capture semantic information that encodes critical context about the transaction. Unlike Lazy Prices [3], which detects textual changes by comparing similarity between discrete time periods, this study introduces a predictive approach: a multi-task time-series LSTM learns an insider’s historical trading and footnote patterns and predicts the expected embedding for the next transaction, capturing temporal anomalies that static comparison cannot detect.

Material differences are identified through dynamic, window-specific percentile thresholds calculated from training data, eliminating manual threshold tuning. The methodology integrates with standard finance event study frameworks [6], demonstrating that detected material changes are associated with significantly lower subsequent abnormal returns (-10.9% CAAR difference at sequence length 128), providing empirical evidence that the model captures meaningful information-driven trading signals not immediately reflected in market prices.

3.2 Model Architecture

The all-MiniLM-L6-v2 sentence transformer model was fine-tuned on domain-specific Form 4 footnote text using contrastive learning with multiple negatives ranking loss. Similar pairs were created by grouping footnotes by issuer (CIK), with up to 2-3 pairs per anchor footnote. The model was selected based on prior research demonstrating that domain-specific tuning on small models can be as performant

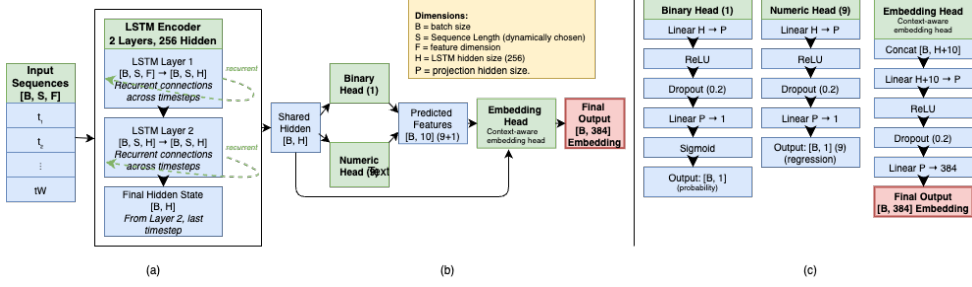


Fig. 1. Proposed LSTM Multi-Task Architecture: From left to right (a) SEC form footnotes are first encoded into semantic embeddings and combined with transactional metadata to form input sequences $[B, S, F]$. These sequences are then processed by a 2-Layer LSTM encoder with 256 hidden units (H), which use recurrent connections to capture temporal data across time-steps. The final hidden layer represents the insider’s past history; (b) Multi-task prediction heads forecast numeric features, 10b5-1 status, and context-aware footnote embeddings, with predicted features concatenated to the hidden state to inform the embedding prediction; and (c) Details of the multi-task prediction heads.

as larger models [8], and its compact size (22.7 million parameters producing 384-dimensional embeddings) enabled training on consumer hardware. Fine-tuning used 2 epochs, learning rate 2×10^{-5} , batch size 64, and 100 warmup steps with the AdamW optimizer. Due to survivorship bias in Tiingo price data prior to 2014, all pre-2014 data was treated as bootstrap training data, with transactions prior to 2012 used for fine-tuning.

These fine-tuned embeddings serve as input features to the LSTM model, which is a multi-task predictor sharing a single 2-layer LSTM encoder (256 hidden units per layer, 0.2 dropout) across all prediction tasks (Fig. 1). The 256 hidden dimension matches the projection head size. The multi-task framework simultaneously predicts transaction features and footnote embeddings, with predicted transaction features providing contextual information that helps the embedding prediction head learn temporal patterns. The LSTM hidden state is concatenated with predicted transaction features (10 dimensions) to form the 266-dimensional input to the embedding prediction head. The prediction heads consist of nine numeric regression heads, one binary classification head, and one context-aware embedding prediction head. The complete model contains 2,022,538 parameters (Table 2). Training used batch size 64, learning rate 1×10^{-3} , weight decay 1×10^{-5} , and the Adam optimizer with default PyTorch parameters.

3.3 Data Preparation and Feature Engineering

For the 2014 and later data, training (Q1’14–Q1’21) and validation (Q1’21–Q1’25) sets contain 458,786 aggregated sales transactions (61,269 10b5-1 plan transactions). Features were normalized using min-max scaling to the $[0, 1]$ range. Input sequences were created using fixed-size sliding windows (stride 1) over transactions grouped by issuer and owner, ordered by transaction date and filing order. This allows for a sliding window based approach to identify transactions for measuring cumulative abnormal returns (CAR).

For example, with sequence size 8, the first 8 transactions predict the footnote embedding for the 9th transaction. Each issuer-insider pair requires at least $W + 1$

transactions to generate a prediction window of size W , so larger window sizes result in fewer qualifying pairs (Fig. 2). Models were trained on sequence sizes of 8, 16, 32, 64, and 128.

Transaction dates were encoded using cosine and sine transformations for each quarter and year. LSTM inputs include: total shares, total value, days since prior sale, days filing was late, cyclical time features (day of year/quarter sine/cosine), 10b5-1 transaction indicator, related footnote shares, and footnote embeddings. The model predicts all input features simultaneously; predicting transaction features alongside embeddings helps the model learn temporal patterns that inform embedding prediction, effectively providing regularization through shared representations.

3.4 Training Procedure

For the 2014 and later models, each insider transaction sequence window size was trained for 100 epochs, using a patience of 10 as an early stopping mechanism. Total loss combines MSE for numeric features, BCE for binary classification (10b5-1 indicator), and cosine similarity loss for embedding prediction:

$$\mathcal{L}_{\text{total}} = \sum_{f \in \mathcal{F}_{\text{num}}} \mathcal{L}_{\text{MSE}}(f) + \mathcal{L}_{\text{BCE}}(10b5-1) + \mathcal{L}_{\text{cosine}}(\text{embedding}) \quad (1)$$

where $\mathcal{F}_{\text{num}}$ includes 9 numeric features (total shares, total value, days since prior sale, days filing was late, cyclical time features, and related footnote shares), and all tasks use equal weighting ($\lambda = 1.0$). Material differences were classified using dynamic, window-specific thresholds calculated as the 15th percentile of the cosine similarity distribution from the training predictions for each window size, representing the most substantial deviations from expected patterns.

3.5 Event Study Framework

Event studies use three windows: estimation (252 trading days), buffer (21 days), and observation (63 days) periods. For each individual event, daily abnormal returns (AR) are calculated as the difference between actual returns and expected returns predicted by the market model during the estimation period. These daily abnormal returns are then cumulatively summed over the observation window to produce cumulative abnormal returns (CAR) for each event. CAAR aggregates these cumulative abnormal returns across all events by computing the weighted average of abnormal returns for each day in the observation window, then cumulatively summing these daily averages. The buffer window separates the estimation period from the observation period to prevent contamination of the model parameters by event-related price variations. To assess the statistical significance of the results, a bootstrap procedure with 1,000 iterations is employed to calculate 95% confidence intervals for the Mean CAR across all market models. The Mann-Whitney U test [14] is also used to determine whether CAAR distributions between material and non-material events are statistically different. This non-parametric test was selected because stock returns exhibit heavy-tailed distributions that violate normality assumptions required by parametric tests such as the t-test.

Table 2. Model Parameter Count Summary

Component	Parameters	Percentage
LSTM Encoder (2 layers)		
Layer 1: Input-to-Hidden	404,480	20.0%
Layer 1: Hidden-to-Hidden	262,144	13.0%
Layer 1: Bias (bias_ih)	1,024	0.1%
Layer 1: Bias (bias_hh)	1,024	0.1%
<i>Layer 1 Subtotal</i>	<i>668,672</i>	<i>33.1%</i>
Layer 2: Input-to-Hidden	262,144	13.0%
Layer 2: Hidden-to-Hidden	262,144	13.0%
Layer 2: Bias (bias_ih)	1,024	0.1%
Layer 2: Bias (bias_hh)	1,024	0.1%
<i>Layer 2 Subtotal</i>	<i>526,336</i>	<i>26.0%</i>
LSTM Total	1,195,008	59.1%
Prediction Heads		
<i>Numeric Regression Heads (9 heads)</i>		
Per head: Linear(256→256)	65,792	3.3%
Per head: Linear(256→1)	257	0.0%
<i>Per head subtotal</i>	<i>66,049</i>	<i>3.3%</i>
9 heads total	594,441	29.4%
<i>Binary Classification Head (1 head)</i>		
Linear(256→256)	65,792	3.3%
Linear(256→1)	257	0.0%
Binary head total	66,049	3.3%
<i>Embedding Prediction Head (1 head)</i>		
Linear(266→256)	68,352	3.4%
Linear(256→384)	98,688	4.9%
Embedding head total	167,040	8.3%
Prediction Heads Total	827,530	40.9%
Model Total	2,022,538	100.0%

4 Experiments

4.1 Benchmark Data

Insider trading data was sourced from the SEC’s SEC Markets Data - Insider Transactions Data Sets (Q1’06–Q1’25) and stock returns from the Tiingo Stock Market API. This analysis uses only SEC Form 4 filings [17]; Form 3, Form 5, and Form 4 amended filings were excluded. Transactions were filtered to tickers matching Tiingo’s supported symbols, with ticker normalization and sales transactions filtered to exclude non-common-stock securities. Daily log returns were calculated as $\ln(\text{adjClose}_t / \text{adjClose}_{t-1})$. The final dataset contains 3,299,234 filings with 6,996,970 total transactions (2,630,559 derivative transactions).

Due to Tiingo’s limitation of providing delisted security history only since 2014, transactions prior to 2014 were used for embedding fine-tuning (1,591,226 sale transactions), while transactions 2014+ were used for training (645,795) and validation (177,563). The training and validation dataset consists of aggregated sales transactions from 2014 onwards, split into training (Q1’14–Q1’21) and validation (Q2’21–Q3’25) sets. Each window size requires at least $W + 1$ transactions per issuer-insider pair to generate predictions, where W is the window size. As window size increases, fewer pairs qualify: 13,699 pairs (21.9%) qualify for window size 8, while only 161 pairs (0.26%) qualify for window size 128 (Table 3).

Regular and derivative securities tables were combined into a single transactions table. Footnotes from all 19 columns are merged into a single array, with null values replaced by [NOF] tokens and converted to pipe-delimited strings for LSTM input. Transactions were aggregated when they shared the same filing, issuer, reporting

Table 3. Event Study Results: Market Model Breakdown and Statistical Significance**Panel A: Detailed Market Model Breakdown by Sequence (CAAR)**

Seq Len	Group	Events	Mkt Mod CAAR	Mkt Adj CAAR	FF3 CAAR	FF5 CAAR	Carhart CAAR	Mean CAAR	Δ Baseline Mean	Avg Win Ratio	Δ Baseline Win	Mat - Non-Mat
Baseline	All	16,077	-4.7%	-7.1%	-4.2%	-4.2%	-4.7%	-4.7%	—	40.8%	—	—
Baseline	Mat	4,586	-4.5%	-7.5%	-4.7%	-4.9%	-5.3%	-5.0%	—	40.2%	—	—
Baseline	Non-Mat	11,492	-4.9%	-6.6%	-3.7%	-3.4%	-4.1%	-4.5%	—	41.1%	—	-0.4%
8	All	23,220	-4.6%	-7.3%	-4.1%	-3.9%	-4.6%	-5.2%	+0.4%	41.1%	-0.3%	—
8	Mat	3,107	-4.8%	-5.4%	-3.8%	-3.4%	-4.2%	-4.3%	-0.7%	39.1%	+1.2%	—
8	Non-Mat	20,113	-4.4%	-9.1%	-4.4%	-4.3%	-4.9%	-5.4%	+0.9%	41.3%	-0.3%	+1.1%
16	All	12,865	-3.6%	-6.0%	-4.6%	-4.0%	-5.0%	-4.2%	-0.6%	42.1%	-1.3%	—
16	Mat	1,690	-5.0%	-4.0%	-5.7%	-4.9%	-6.2%	-5.2%	+0.2%	35.5%	+4.7%	—
16	Non-Mat	11,175	-2.9%	-8.0%	-3.4%	-3.1%	-3.8%	-4.2%	-0.3%	42.9%	-1.9%	-0.9%
32	All	6,399	-1.5%	-4.4%	-2.4%	-1.6%	-2.7%	-1.3%	-3.5%	46.2%	-5.5%	—
32	Mat	840	-4.1%	-2.5%	-4.3%	-3.2%	-4.5%	-3.7%	-1.3%	37.6%	+2.6%	—
32	Non-Mat	5,559	1.1%	-6.2%	-0.4%	0.1%	-0.9%	-1.3%	-3.3%	47.3%	-6.2%	-2.5%
64	All	2,840	0.1%	-3.4%	-1.8%	0.4%	-2.9%	2.8%	-7.5%	50.2%	-9.4%	—
64	Mat	336	-7.7%	-3.7%	-8.1%	-5.1%	-9.0%	-6.7%	+1.8%	32.3%	+7.9%	—
64	Non-Mat	2,504	7.9%	-3.0%	4.5%	5.9%	3.3%	3.7%	-8.2%	52.3%	-11.2%	-10.4%
128	All	897	4.5%	-2.8%	0.9%	-1.3%	0.3%	3.6%	-8.3%	51.1%	-10.4%	—
128	Mat	165	-1.9%	-5.9%	-4.8%	-8.3%	-4.7%	-5.1%	+0.2%	32.4%	+7.8%	—
128	Non-Mat	732	10.8%	0.4%	6.5%	5.8%	5.3%	5.3%	-10.3%	55.5%	-14.5%	-10.9%
All	All	9,244	-0.1%	-4.8%	-2.0%	-1.3%	-2.4%	-0.9%	-3.9%	46.1%	-5.4%	—
All	Mat	1,228	-4.8%	-4.6%	-5.2%	-4.7%	-5.1%	-5.0%	+0.0%	35.4%	+4.8%	—
All	Non-Mat	8,017	1.0%	-4.8%	-1.0%	-0.4%	-1.4%	-0.3%	-4.3%	47.9%	-6.8%	-4.7%

Panel B: Aggregated Means with 95% Bootstrap Confidence Intervals

Seq	Group	Mean CAR (B)		Mean CAR (M)		Δ Mean (B-M)	Count (B)	Count (M)	Win (B)	Win (M)	Δ Win (B-M)
		[95% CI]	[95% CI]	[95% CI]	[95% CI]						
8	All	-4.8%	[-5.1%, -4.5%]	-5.2%	[-5.5%, -4.8%]	+0.3%	23,220	23,220	40.8%	41.1%	-0.3%
8	Material	-5.4%	[-6.0%, -4.7%]	-4.3%	[-5.1%, -3.5%]	-1.1%	3,107	3,107	40.1%	39.1%	+1.0%
8	Non-material	-4.5%	[-4.9%, -4.1%]	-5.4%	[-5.8%, -5.1%]	+0.9%	20,113	20,113	41.1%	41.3%	-0.2%
16	All	-4.5%	[-5.0%, -3.9%]	-4.2%	[-4.6%, -3.7%]	-0.3%	12,865	12,865	41.4%	42.1%	-0.7%
16	Material	-5.9%	[-7.1%, -4.8%]	-5.2%	[-6.2%, -4.1%]	-0.8%	1,690	1,690	38.0%	35.5%	+2.6%
16	Non-material	-3.9%	[-4.5%, -3.4%]	-4.2%	[-4.7%, -3.7%]	+0.3%	11,175	11,175	42.6%	42.9%	-0.4%
32	All	-2.0%	[-2.7%, -1.4%]	-1.3%	[-1.9%, -0.6%]	-0.8%	6,399	6,399	45.0%	46.2%	-1.2%
32	Material	-6.3%	[-8.1%, -4.6%]	-3.7%	[-5.0%, -2.4%]	-2.6%	840	840	35.0%	37.6%	-2.6%
32	Non-material	-1.2%	[-1.9%, -0.5%]	-1.3%	[-1.9%, -0.6%]	0.0%	5,559	5,559	47.1%	47.3%	-0.2%
64	All	-0.1%	[-0.9%, 0.6%]	2.8%	[1.9%, 3.8%]	-2.9%	2,840	2,840	46.5%	50.2%	-3.7%
64	Material	-6.7%	[-9.4%, -4.0%]	-6.7%	[-8.5%, -5.1%]	0.0%	336	336	27.3%	32.3%	-5.0%
64	Non-material	0.9%	[0.2%, 1.7%]	3.7%	[2.7%, 4.7%]	-2.8%	2,504	2,504	49.8%	52.3%	-2.5%
128	All	-4.0%	[-5.0%, -2.9%]	3.6%	[2.4%, 4.7%]	-7.5%	897	897	32.6%	51.1%	-18.6%
128	Material	-3.4%	[-6.9%, 0.1%]	-5.1%	[-7.8%, -2.5%]	+1.8%	165	165	36.3%	32.4%	+3.9%
128	Non-material	-4.1%	[-5.1%, -3.1%]	5.8%	[4.5%, 7.1%]	-9.8%	732	732	31.5%	55.5%	-24.1%

owner, transaction date, transaction code, and derivative status. Transactions are matched to stock returns using ticker symbol and transaction date.

Two distinguishing transaction features require special handling: SEC Rule 10b5-1 trading plans [10,18] and derivative security conversions. 10b5-1 plans are identified by parsing footnote text for Rule 10b5-1 language. For derivative conversions, M code transactions (option exercises) are matched to subsequent S or D code transactions (sales or disposals) within the same Form 4 filing using sequential windowing. Footnotes from linked M transactions are aggregated into `related_footnotes`, and primary and related footnotes are combined with a `[REL]` separator token.

4.2 Baseline Method

The baseline method computes L2-normalized mean footnote embeddings for each issuer-insider pair (cik, rptcik) from all training data, creating a time-agnostic centroid as the expected footnote representation. Validation footnotes are compared using cosine similarity. The baseline requires a minimum of three training transactions

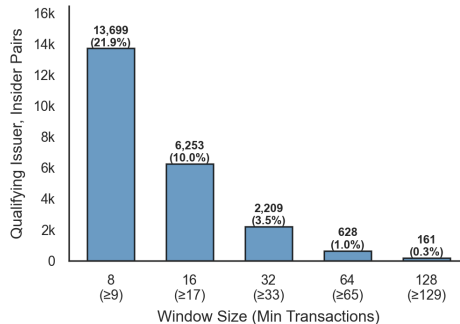


Fig. 2. Number of qualifying issuer-insider pairs by window size. Each window requires at least $W + 1$ transactions per pair, where W is the window size.

Table 4. Embedding Model Cosine Similarity Comparison

Model	Emb Dim	Silhouette	Same Issuer	Diff Issuer	Difference
footnote_embeddings	384	0.3027	0.8268	0.4835	0.3433
fin-emb-investopedia	768	-0.0150	0.7933	0.4883	0.3050
all-MiniLM-L6-v2	384	-0.0217	0.7650	0.4638	0.3012
all-mpnet-base-v2	768	-0.0273	0.8036	0.5535	0.2501

per pair and uses per-pair materiality thresholds (bottom 15% of cosine similarity), resulting in 19,113 material events (27.54% material difference rate) compared to 177–3,552 events for multi-task models (11–18% rates) (Table 6). While the static approach lacks temporal context and cannot adapt to evolving footnote patterns, it provides adequate sample sizes for event study analysis (Table 3).

4.3 Implementation Details

All experiments were conducted on an Apple M4 Max MacBook Pro. The implementation used Python with PyTorch, NumPy, Pandas, and Transformers for model development, PostgreSQL and DuckDB for data processing, and the eventstudies library [1] for event study analysis. Data normalization combined non-derivative and derivative transactions into a single table with footnote text, exported to Parquet format for aggregation and filtering. Model training used PyTorch’s MPS GPU library for Apple Silicon acceleration. All pretrained models were downloaded from Hugging Face and executed locally.

4.4 Embedding Model Evaluation

The sentence transformer model was fine-tuned on Form 4 footnote text for 2 epochs. An evaluation comparing the fine-tuned model against other pretrained models using Silhouette scores (based on cosine similarity) showed the fine-tuned model achieved the best cluster separation (Silhouette score 0.3027, with scores > 0 indicating clearly defined clusters) (Table 4). With 384-dimensional output, it achieves the largest difference between average cosine similarity for same-issuer footnotes (0.8268) and different-issuer footnotes (0.4835) among all models compared.

Table 5. Multi-Task LSTM Model Prediction by Sequence

Seq	Split	Samples	Cos Sim	$\pm$ Std	RMSE	R ²
8	train	164,075	0.7718	0.0886	0.0345	0.5434
8	val	26,212	0.7678	0.0818	0.0348	0.5355
16	train	101,618	0.7704	0.0887	0.0346	0.5407
16	val	14,688	0.7664	0.0797	0.0349	0.5327
32	train	53,427	0.7720	0.0868	0.0345	0.5440
32	val	7,330	0.7661	0.0724	0.0349	0.5322
64	train	24,657	0.7747	0.0839	0.0343	0.5493
64	val	3,227	0.7656	0.0605	0.0349	0.5312
128	train	9,221	0.7866	0.0736	0.0333	0.5730
128	val	986	0.7697	0.0541	0.0346	0.5393

Table 6. Baseline vs Multi-Task Model Comparison: Similarity Distributions and Material Difference Rates

Sequence Length	Training Cosine Similarity	Validation Cosine Similarity	Shift	Material Rate	Material Events
Baseline	0.8820	0.8422	−0.0398	27.54%	19,113
8	0.7718	0.7678	−0.0040	13.55%	3,552
16	0.7704	0.7664	−0.0040	13.26%	1,948
32	0.7720	0.7661	−0.0059	12.70%	931
64	0.7747	0.7656	−0.0091	10.88%	351
128	0.7866	0.7697	−0.0169	17.95%	177

4.5 Multi-Task Method

Bootstrap analysis used fine-tuning period data (2006–2011 training, 2012–2013 validation) to form transaction sequences of window sizes 8, 16, 32, 64, and 128. Models were trained for 100 epochs with a patience of 10. Initial threshold estimation used mean minus one standard deviation of cosine similarity scores, yielding an average of 0.65. For 2014+ data, percentile-based thresholds from training data were used for each window size, with the 15th percentile selected to capture the bottom 15% while maintaining alignment with bootstrap findings.

Regression analysis shows all models achieve mean cosine similarity of approximately 0.77, with standard deviation decreasing as window size increases, RMSE remaining approximately 0.034, and R² averaging 0.53 (Table 5). Compared to the baseline (Table 6), the multi-task model shows sequence-specific similarity scores (0.77–0.79 training, 0.77–0.77 validation) with minimal shifts (−0.004 to −0.017), while the baseline uses static centroids across all sequence lengths (0.8820 training, 0.8422 validation, shift −0.0398). The baseline’s static approach produces a material difference rate of 27.54% (19,113 events) compared to 11–18% for multi-task models (177–3,552 events per sequence) (Table 6).

4.6 Event Study Results

Event studies were conducted for sequence windows of 8, 16, 32, 64, and 128 using estimation (252 trading days), buffer (21 trading days), and observation (63 trading days) periods starting at trade date + 1. Five market models estimate abnormal returns: Market Model and Market Adjusted Model (both using S&P500 ETF (SPY) as benchmark), and Fama-French 3, Fama-French 5, and Carhart models. Material events are defined as those in the bottom 15% of cosine similarity (15th percentile threshold from training data) for each window size. Returns outside ± 3.5 Median Absolute Deviations (MAD) from the median were removed as outliers.

Mann-Whitney U tests determine whether CAAR distributions between material and non-material events are statistically different. The expected pattern, where material footnote changes are associated with more negative CAAR than non-material changes, does not consistently emerge for shorter sequence lengths. For sequence lengths 8, 16, and 32 (Table 3, Panel A), the Market Adjusted Model exhibits the opposite pattern, with non-material events showing more negative CAAR than material events (-9.1% vs -5.4% at length 8, -8.0% vs -4.0% at length 16, -6.2% vs -2.5% at length 32). The expected pattern emerges consistently across all market models starting at sequence length 64, suggesting that shorter sequence lengths do not provide sufficient historical context for the model to learn patterns that align with subsequent return behavior.

Sequence lengths 64 and 128 perform well across all models (Table 3, Panel A). Averaging across all five market models, material events show declining mean CAAR (from -4.3% at length 8 to -5.1% at length 128) and win ratio (from 39.1% to 32.4%), while non-material events show increasing mean CAAR (from -5.4% to $+5.8\%$) and win ratio (from 41.3% to 55.5%). Win ratio (the proportion of events with positive abnormal returns) measures trading performance; for sales transactions, lower win ratios indicate better performance. The diverging trends as sequence length increases demonstrate that longer sequences enable the model to better distinguish material from non-material changes. The statistical significance of these results is reinforced by the non-overlapping 95% bootstrap confidence intervals in Panel B of Table 3. For sequence length 128, the Mean CAR (M) interval for material changes ($[-0.0782, -0.0247]$) is entirely separated from the interval for non-material changes ($[0.0449, 0.0705]$), providing robust evidence that the model identifies meaningful signals.

The baseline model uses static issuer-insider pair centroids, resulting in higher similarity scores (training: 0.8820, validation: 0.8422) and a negative shift of -0.0398 (Table 6). In contrast, the multi-task model shows minimal train-validation distribution shift (shifts of -0.004 to -0.017). The baseline produces a material difference rate of 27.54% (19,113 events), higher than the multi-task model’s 11–18% rates (177–3,552 events per sequence). However, the baseline’s static centroid approach lacks temporal context and cannot adapt to evolving footnote patterns. The multi-task architecture leverages temporal context to enable more nuanced detection of material changes that are more predictive of abnormal returns (Table 3).

4.7 Threshold Sensitivity Analysis

To verify that results are not sensitive to the choice of similarity threshold, we re-ran the event study across percentile levels from 5% to 30% for sequence windows 64 and 128, averaging CAAR across all five market models (outliers excluded), consistent with the methodology of Table 3. The 5th percentile was excluded from Table 7 due to insufficient material event counts ($n \leq 40$ for Window 64, $n \leq 37$ for Window 128), yielding statistically unreliable estimates. Table 7 reports mean CAAR for material and non-material groups and the resulting signaling gap across the reliable 10%–30% range. Across all tested thresholds, the gap remains consistently negative and statistically significant (Mann-Whitney U, $p < 0.05$), ranging from -6.1% to -10.4% for Window 64 and from -6.7% to -13.0% for Window 128. The 15th

Table 7. Threshold Sensitivity: Mean CAAR averaged across all five market models (outliers excluded)

Pct	Window 64				Window 128			
	n_{mat}	CAAR _{mat}	CAAR _{non}	Gap	n_{mat}	CAAR _{mat}	CAAR _{non}	Gap
10%	163	-6.0%	+3.1%	-9.1%	91	-2.3%	+4.4%	-6.7%
15%	339	-6.7%	+3.7%	-10.4%	165	-5.1%	+5.8%	-10.9%
20%	503	-5.5%	+3.9%	-9.4%	217	-6.0%	+7.0%	-13.0%
25%	629	-3.2%	+3.9%	-7.1%	232	-5.8%	+7.1%	-12.9%
30%	737	-2.2%	+3.9%	-6.1%	283	-3.1%	+7.1%	-10.2%

percentile used in the main analysis yields the strongest signaling gap for Window 64 (-10.4%) and near-optimal results for Window 128 (-10.9%), indicating that this threshold is not a conservative convention but the empirically best-supported choice for the primary benchmark.

5 Conclusions

In this paper, we study the effectiveness of deep learning approaches for identifying material changes in SEC Form 4 footnote disclosures for AI driven trading behavior modeling and monitoring. By fine-tuning a sentence transformer model on Form 4 footnote text and employing a multi-task LSTM architecture to predict footnote embeddings from historical transaction patterns, the study processes over 3.3 million Form 4 filings from 2006 through 2025. The multi-task architecture effectively predicts footnote embeddings with mean cosine similarity of approximately 0.77 and R^2 averaging 0.53 across all sequence lengths, demonstrating that the model learns meaningful patterns from historical transaction sequences.

Event study results show that the multi-task LSTM model significantly outperforms the static centroid baseline, with significant improvements in CAAR and win ratio separation at longer sequence lengths (64 and 128). The CAAR difference (material minus non-material) increases from 1.1% at sequence length 8 to -10.9% at sequence length 128, and the separation is statistically robust. The baseline’s static centroid approach treats all historical footnotes equally regardless of when they occurred, lacking the temporal specificity to capture evolving patterns. Multi-task LSTM’s ability to learn an expected footnote for a specific insider at a specific point in their transaction history enables more precise detection of meaningful deviations.

Future work could expand beyond sales transactions to incorporate multiple transaction codes, enhance footnote aggregation accuracy for derivative security conversions, incorporate holdings changes and security title filtering, and extend the analysis to Form 3 and Form 5 filings to provide a more comprehensive view of insider trading patterns.

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